

Exoplanetary Rover SLAM

Problem ID: problem

A fleet of autonomous rovers is exploring a subterranean cave system on exoplanet Kepler-16b. Each rover runs a SLAM (Simultaneous Localisation and Mapping) algorithm tracking N distinct geological landmarks, maintained in an $N \times N$ state transition matrix. During the latest exploration phase, each rover recorded M sequential state transitions $M_1, M_2, \dots, M_M$, capturing how the map evolved step by step. The navigation module, which is prone to memory corruption from the cave's intense magnetic interference, has produced a matrix E representing its current belief of the map state after applying all transitions in sequence. The rovers use E to plan their next move and if it is wrong, they are navigating on a corrupted map and risk driving straight into a cave wall.

To avoid overflow in the processors of the rover, all matrix computations are performed modulo

$$P = 10007.$$

Before giving the rovers the green light, you must verify from ground control that

$$E \equiv M_1 \times M_2 \times \dots \times M_M \pmod{P}.$$

Every second the rovers sit idle drains their battery, so you need an answer fast. Transmitting full matrices is too slow, so each rover instead transmits only the seed used to generate each transition matrix. You reconstruct them locally using a shared PRNG. Given a seed S , let $x = S$. The entries of the matrix are filled in row-major order with rows and columns indexed from 0. For each entry:

$$x = (911382323 \cdot x + 972663749) \bmod P$$

$$M_{ij} = x$$

Input

The first line contains an integer $1 \leq N \leq 400$, the matrix dimension. The second line contains an integer $1 \leq M \leq 60$, the number of transition matrices. The following M lines each contain a single integer $1 \leq S_i \leq 10^9$, the seed for matrix M_i . It is guaranteed that $S_i \neq S_j$ for all $i \neq j$. The final N lines each contain N space-separated integers describing the current map state matrix E . Each entry satisfies $0 \leq E_{ij} < P$.

Output

Output YES if the rovers can safely advance, otherwise output NO.

Scoring

Your solution will be tested on a set of test groups, each worth a number of points. To get the points for a test group you need to solve all test cases in the test group.

Group	Points	Constraints
1	20	$N \leq 100$ and $M \leq 5$
2	80	No additional constraints

Sample Input 1

4 2 926756583 911666163 971 8952 3646 6152 6436 8548 9610 8950 8378 5225 5658 4767 5675 9474 894 2834	NO
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Sample Input 2

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4
2
144272510
611178003
757 4278 846 1622
4722 7216 2500 7975
9370 1014 8525 3631
6384 9399 6338 8523
```

Sample Output 2

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YES
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